1. Chained-S3 SAT on N83, m = 5…9: meet-in-the-middle lookup for the middle intermediate

Run `n83_chain_sat_meet_20261101`. m = 5, 6, 7: 2026-11-01 01:50:17Z – 02:04:52Z, protocol `PROTOCOL.md` frozen at `0704c3b008`. m = 8, 9: 02:11:51Z – 02:47:51Z, amendment `PROTOCOL_m8_9.md` frozen before its data (user request). Driver `examples/n83_chain_sat_bench.rs`; native analysis `examples/n83_chain_sat_analyze.rs` ⟶ `ANALYSIS.json`, `table.tsv`; graphs by `plot.py` from `ANALYSIS.json` only. Host: one 4-vCPU x86-64 cloud VM, L0, one core per m. Follow-up to `research/n83_chain_sat_order_20261031`, where every arm was wall-capped at m ≥ 5.

Curve. `icv1-f2m83-t6151469093347-cdcc5432` (`KoblitzCurve::known_n83_k1`), r = 8569786107849059. Base: Frobenius union of span(1, z), 167 points (closed under negation). Same seed and targets as the 10-31 run: 3 public $[a]G$ and 2 planted sums of m base points per m; per query 300,000 conflicts and 120 s.

1.1. The lookup (`ChainS3Tables::meet = d`)

The middle intermediate y_{d-1} is the x-coordinate of a signed sum of d base points, and of R minus a signed sum of the other $m - d$. The x-fingerprints of every signed d-sum are built once per base (batched inversion, query-independent, reported as warmup); per query every signed $(m - d)$-sum s gives $R - s$, kept when its x is in that set. The survivors are the exact domain of y_{d-1} (a fingerprint collision can only add a value). Patterns with the identity at y_{d-1} need R itself to be a signed $(m - d)$-sum and are refuted without a solve otherwise; all other patterns are refuted without a solve when the domain is empty. Unit tests: batched additions equal `point_add`; the depth-2 set equals the exact sum table; a planted witness is kept; a target with a zero-sum decomposition is never refuted.

1.2. Results (measured)

m	d	warmup (once)	A2 control: public / planted	A4 lookup: public refuted	A4 median public s	A4 planted domain sizes
5	3	0.68 s, 347,688 fp	0/3, 0/2	**3/3** at 0 conflicts	0.80	10, 16
6	3	0.68 s	0/3, 0/2	**3/3** at 0 conflicts	1.39	20, 62
7	4	26.1 s, 12,976,303 fp	0/3, 0/2	**3/3** at 0 conflicts	2.54	35, 35
8	4	27.2 s	0/3, 0/2	**3/3** at 0 conflicts	76.7	70, 250
9	5	958.8 s, 372,406,144 fp	0/3, 0/2	0/3 (120 s cap)	—	—

1. **Soundness.** 50 records, 0 soundness flags. No planted target refuted by either arm.
2. **Public targets, m = 5…8.** The lookup arm refutes all 12 with an empty domain and 0 solver conflicts; the control (ordering + depth-2 tables, the best arm of the 10-31 run) decides none in 120 s. Per the frozen rule this is an improvement at m = 5, 6, 7 and 8 (more decided public queries, no planted target lost).
3. **Planted targets.** Neither arm finds a witness within 120 s at any m ≥ 5. The lookup domain is non-empty on every planted target (10–250 values) and empty on every public one: the lookup separates them, but the SAT search over the rest of the chain does not finish within the cap.
4. **m = 9.** The depth-5 set (372M fingerprints, 959 s) is built, but each query's back enumeration (≈ 3.4·10^7 additions, each with a binary search in a 3 GB sorted array) exceeds 120 s: no improvement at m = 9.

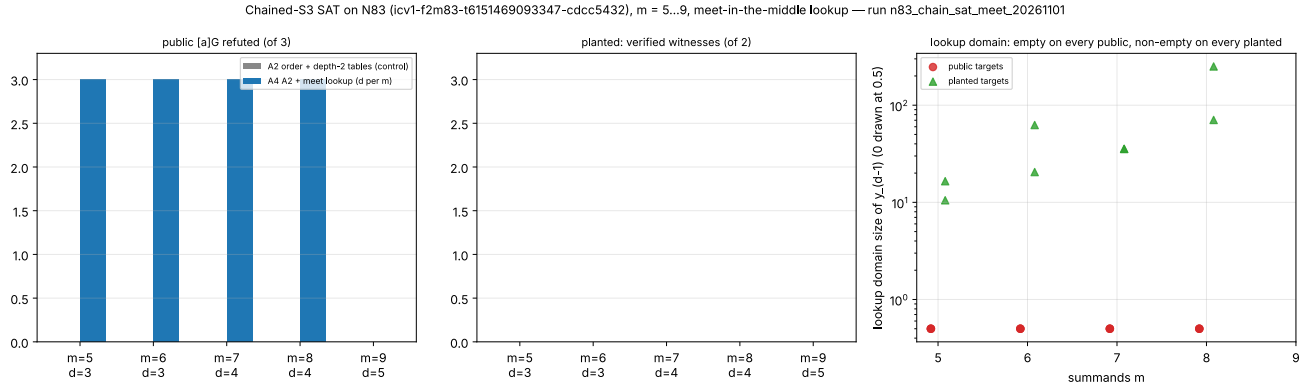


Figure 1: Public queries refuted, planted witnesses verified, and the lookup domain size of the middle intermediate per target. m = 9 has no domain: its enumeration hit the cap.

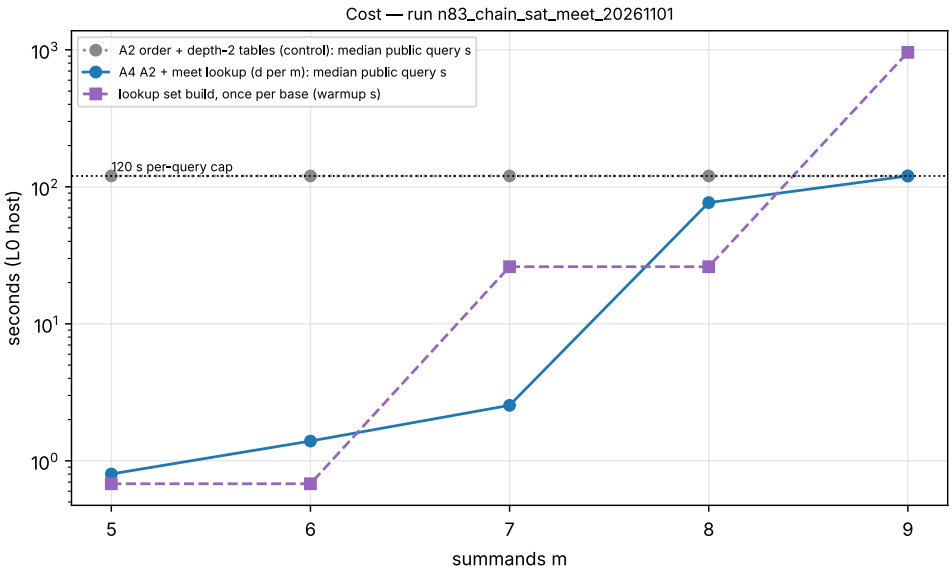


Figure 2: Median public query seconds per arm and the once-per-base lookup build (L0 host).

1.3. What this is and is not

- The refutations are a meet-in-the-middle over the factor base placed in front of the SAT instance: an empty domain proves no decomposition exists, without search. Per-query cost grows as $|B|^{m-d}$ with a $|B|^{d}$ precomputation, which is the usual MITM trade-off, not a sub-exponential step.
- Decomposition stage only: no relations were collected and no DLP solved. No leaderboard or `benchmarks/` row (validation gate: no replay certificate, validator report or matched-rho `VALIDATION.json`). Matched rho on this curve: 1.51 s mean online (`research/n83_msweep_20261024`), so the distance remaining is unchanged in kind.

1.4. Deviations and limits

- `PROTOCOL.md` priced the lookup with 334 signed points; the base is already negation-closed (167), so costs are $\approx 2^{d}$ smaller than declared. m = 8, 9 were added by a separate frozen amendment rather than by editing the protocol.
- L0 host, 5 targets per m, one base size; m = 8 and 9 ran in parallel and the m = 9 queries overlapped the m = 8 run.
- EKS (AGENTS.md §9a) not used: not an ecbench workload, no aws CLI in this session.
- Next levers (not measured): reconstruct the planted witness directly from the lookup hit (re-enumerate the matching d-multiset) instead of leaving it to SAT; a direct-indexed hash in place of binary search for the m = 8, 9 lookups.